YOUTUBE TRENDING INTELLIGENCE

Project 4 — YouTube Trending Video Analytics

Cleaned records analyzed	Countries	Tools
112,735	India · USA · Great Britain	Python · Pandas · Matplotlib · SQLite/SQL

1. Introduction

This project explores historical YouTube trending-video records to identify patterns in video categories, views, likes, comments, tags, title length, and audience engagement across India, the United States, and Great Britain. The dataset contains daily records of videos that appeared on the trending list; it is not a complete record of every YouTube video or all-time most-viewed videos.

2. Abstract

Three regional datasets were cleaned and analyzed using Python and SQL. Duplicate records were removed, missing descriptions were filled with empty text, and category IDs were mapped to category names using the provided JSON files. Additional fields such as engagement rate, title length, tag count, and time-to-trend were created for exploration. Visualizations were generated to compare category performance and content characteristics. SQL queries were used to summarize categories, views, engagement, and cross-country patterns.

3. Tools Used

- Python and Jupyter Notebook — analysis workflow.
- Pandas — data cleaning, transformation, and aggregation.
- Matplotlib — charts and visual comparisons.
- SQLite and SQL — queries across the three country tables.
- CSV and JSON files — regional data and category-name mapping.

4. Steps Involved

- Loaded India, USA, and Great Britain CSV datasets and their category JSON files.
- Removed duplicate rows and filled missing description values; saved cleaned CSV files.
- Added analysis fields including category name, engagement rate, title length, tag count, and trend timing.
- Explored category counts, average views, likes, dislikes, comments, title length, tag counts, and trending activity.
- Stored cleaned datasets in SQLite tables and wrote SQL queries for category summaries, top videos, and cross-country comparisons.
- Saved charts in the **visualizations** folder and SQL queries in **SQL/youtube_analysis_queries.sql**.

5. Key Findings

Average views in the leading category for each country

Country	Leading category	Average views
India	Gaming	3.61 million
USA	Music	6.20 million
Great Britain	Music	12.44 million

Average engagement rate by country

Country	Average engagement rate
India	2.45%
Great Britain	3.80%
USA	3.89%

- Music was the leading average-view category in the USA and Great Britain, while Gaming led in India.
- Film & Animation appeared among the five highest-average-view categories in all three countries.
- Average engagement rates in the analyzed records were 2.45% for India, 3.80% for Great Britain, and 3.89% for the USA.
- In the USA dataset, Music had the highest average views by category, while Nonprofits & Activism had high average likes and comments. Category averages can be affected by viral outliers.

6. Conclusion

The analysis identifies differences in trending categories and engagement patterns across the three regions. Music was prominent in the USA and Great Britain, while Gaming led by average views in India. The country-level engagement comparison showed the USA highest, followed closely by Great Britain, with India lower in this sample. These findings can help describe historical content trends, but they should not be treated as proof that a category will trend in the future. The analysis is descriptive and limited to the selected historical dataset.

Dataset source

Data source: Kaggle, “Trending YouTube Video Statistics” by DataSnaek. Kaggle dataset page.

Note: Average metrics are calculated from trending records in the cleaned datasets, so repeated appearances of a video may contribute multiple records to category-level averages. Top-video queries used video IDs to reduce repeated-video rows.